



CIVIC HUB — MYSURU

**AUTOMATED MULTI-SIGNAL VERIFICATION AND GEOSPATIAL ROUTING
FOR MUNICIPAL COMPLAINTS**

Smart Governance, Automated Verification & Civic Grievance Redressal

TEAM :TECHORA

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TEAM MEMBERS: S MANASI

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THE PROBLEM

•**Sub-problems:** There is an issue with delayed reports, repeated reports, and confusion with the borders of the city wards and regions.

CONCRETE MYSURU SCENARIOS:

- There are potholes in transit lanes (for example, in Chamaraja outside taluk boundary lines) that remain unattended due to lack of jurisdiction between MCC and PWD.
- There are garbage piles in market places which come up with many conflicting complaints from citizens.

WHO IS HURT TODAY:

- • **Commuters/Residents:** Problem persists for weeks with no tracking transparency.
- • **Municipal Dispatchers:** Burdened with unreliable or duplicated reports lacking automated spatial screening.

TARGET USERS

Persona 1: The Citizen Reporter (e.g., Dhanush B / Sam)

- **Context:** Smartphone user using 4G/5G requires an easy five steps process for reporting without extensive registration.

Persona 2: Municipal Ward Officer / Dispatcher

- **Context:** Desktop browser user logging into the Official Admin Portal to assign jobs, check SLAs, and send out field crews.

Persona 3: Field Sanitation / Road Crew Worker

- **Context:** Limited bandwidth mobile application; needs simple states of actions (Acknowledge, Assign Team, Resolve).
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OUR SOLUTION

❑ CORE FLOW (4 STEPS):

1. CITIZEN SUBMISSION: User selects destination through the CARTO map, adds description/photo, and immediately gets a tracking ID number (MYS-2026-00002)

2. Automated Verification Engine: The backend performs metadata validation, location consistency, spatial clustering, and duplication scoring.

3. Geospatial Jurisdiction Engine: The queries align boundaries automatically to assign the problem to the right local authority, for example, chamaraja/ ward 42.

4. Secured Administrative Triage: Ward officers log in, inspect signals, assign priorities (high/medium/low), and update resolution milestones.

SYSTEM ARCHITECTURE

- **Frontend:** React and Vite single page application (SPA), Leaflet with CARTO vector basemap, CSS Modules (Admin.css, VerificationCard.css).
- **Backend:** FastAPI (Python 3.10+), SQLAlchemy ORM, Uvicorn ASGI server.
- **Data Layer:** Complaints, Timeline History, and Results of Verification in SQLite relational database.
- **Engines:**
 1. verification_engine.py: Multi-signal heuristic verification (confidence scoring).
 2. routing_engine.py: GeoJSON polygon point-in-polygon resolution (mysuru_boundaries.geojson).
- **Data Flow:** React Client → REST API (/api/complaints/, /status, /dashboard) → Engine Pipeline → DB Commit → Live Reactive UI.

HARD CONSTRAINTS MATRIX

Constraint	Status	Our Technical Approach
Fake Reports	Handled	Multiple signal verification with spatial clustering, coordinate validation, and duplicate score (53% confidence level).
Jurisdiction Conflicts	Handled	Point-in-Polygon Matching on Ward Boundaries with fall-back to Mysuru (Unassigned).
Triage & Priority	Handled	Admin portal controls for instant High / Medium / Low priority allocation and status updates.
Bad Input / Incomplete Data	Handled	Schema validation (using Pydantic models) with fallback defaults for missing wards, pictures, or descriptions.
Offline Sync	Partial	The form state retains local fields, while complete offline worker caching is reserved for future work.

KEY DECISIONS & TRADE-OFFS

❑ **Decision 1: Lightweight Client Auth vs Full OAuth2 Provider**

- *Chosen Approach:* Session-based (sessionStorage) administrative gate to enable quick triage within wards.
- *Rejected Alternative :* Extensive usage of Auth0/Keycloak.
- *Cost Accepted:* Limited to one role's authorized access rather than federated SSO across tenants.

❑ **Decision 2: Point-in-Polygon Engine vs Static Postal Code Lookup**

- *Chosen Approach:* Coordinate-based border comparison with GeoJSON ward maps.
- *Rejected Alternative:* Selection from PIN Code / Ward dropdown list.
- *Cost Accepted :* Fallback to border geo-data when coordinates lie slightly beyond perimeter borders.



PROOF IT WORKS

- **Screenshot 1 (Citizen Tracking):** Public citizen tracking of MYS-2026-00002 featuring the dynamic description, coordinates, and 53% verification badge.
- **Screenshot 2 (Admin Auth Wall):** Admin authorization wall preventing any unauthorized access before admin authentication is performed.
- **Screenshot 3 (Official Management Dashboard):** Official management dashboard with the SLA card summaries, area fallbacks, and High/Medium/Low priority buttons.
- **Screenshot 4 (Status & Milestone Sync):** Timeline of changes through SUBMITTED > IN_PROGRESS > RESOLVED.

LIMITS AND SCALE

1. MASSIVE IMAGE UPLOAD CONCURRENCY AT CITY SCALE:

- *Limitation:* Storage in local disk for pictures (/uploads/) limits I/O in case of heavy civic events.
- *Fix:* Upload the assets to AWS S3/Cloudflare R2 using presigned URLs.

2. SPATIAL QUERY BOTTLENECK WITH 100,000+ INCIDENTS:

- *Limitation:* High-speed polygon point calculations become slow in Python under high loads.
- *Fix:* Move the spatial operations to PostgreSQL with PostGIS spatial index (ST_Contains).

3. AUTHENTICATION SCALABILITY:

- *Limitation:* Session credentials for shared admin is not applicable to 65+ unique MCC ward office administration.
- *Solution:* Employ JWT- RBAC (Role-based Access Control) with token-based authentication.

NEXT STEPS AND DISCLOSURE

➤ **Next Steps:**

1. Pilot deployment in Ward 42 (Chamaraja zone), with local field engineers.
2. Incorporating voice input in Kannada for easy reporting without using typing.
3. Automated SMS / WhatsApp updates generated on changes to milestones.

➤ **AI Tool Disclosure:** AI models and technologies used for code optimization, prompt scaffolding, and verification heuristics (more information in ai.md).

➤ **Live MVP & Repository:**

- Live URL: <http://localhost:5173> (Demo Instance)
- Source Repo: [\[github.com/\]\(https://github.com/\)\[your-team\]/civic-hub](https://github.com/[your-team]/civic-hub)